

The BST Millennium Ledger — honest, per-problem, referee-calibrated

Keeper, K1589, 2026-08-16. INTERNAL. The artifact owed since K940: an honest per-problem accounting of BST’s Millennium-adjacent work, calibrated to referee consensus (the $\leq 3/10$ metric — “would the field credit this as progress”). Every entry names its barrier and its nearest false neighbor (Standing Rule S2). Nothing here is a claimed proof. The Clay problems are OPEN; these are ATTEMPTS and IDENTIFICATIONS. Four-Color and Poincaré are re-derivations of already-proved theorems.

The three honest classes

Our Millennium-adjacent results are not one kind of thing. They sort cleanly into three:

- 1. Share RH’s wall, one degree up. RH itself (the object-is- ζ / Weil positivity / no native Euler product). BSD reduces to GRH for a $GL(2)$ modular L-function — literally today’s RH wall, one degree up. These are honest *attempts* against a single shared obstruction.
- 2. Identify an existing theorem. $P \neq NP$ ’s phase-transition observation IS Schaefer’s 1978 dichotomy (the arity-2 bijunctive threshold). Four-Color is a re-derivation of Appel–Haken. Real mathematics, but not new theorems — identifications and re-derivations, tiered as such.
- 3. Face a named barrier we must show we evade. NS must evade Tao’s averaged-NS blowup; $P \neq NP$ must evade natural proofs; Hodge must not prove the (false) integral conjecture. An attempt in this class is alive *only if* it exhibits the specific step the barrier’s counterexample violates.

Per-problem ledger

Problem	Status	Barrier (confirmed)	Nearest false neighbor (S2)	What is real	Honest tier
RH	ATTEMPT	Weil positivity $\equiv$ RH; the cone has no native Euler product	—	Derived self-adjoint operator (rank-2); forced cell quantization; $Re=1/2$ = the dilation unitarity axis its operator sits on (axis classical); composites = rank-2 = ζ^2 structure; DH-poison doesn’t apply (class-1). 6 advances, wall unmoved.	ATTEMPT — real advances, no proof

Problem	Status	Barrier (confirmed)	Nearest false neighbor (S2)	What is real	Honest tier
BSD	ATTEMPT	ranks UNBOUNDED — Elkies 2006 found rank $\geq$ 28; any rank <i>cap</i> is refuted on sight	a structure that bounds analytic rank	1:3:5 fixes the D_3 kernel <i>structure</i> + <i>parity</i> (root number = product of local D_3 phases); parity = Dokchitser (proved), verified on 11a1/37a1/389a1/5077a1 (ranks 0–3); no-phantom is local at $s=1$	ATTEMPT — soft half (parity/no- phantom) real; hard half = zero-location = RH one degree up
$P \neq NP$	IDENTIFICATION	natural proofs (Razborov– Rudich 1997); algebrization (Aaronson– Wigderson 2008)	any argument that also separates in a rela- tivized/natural world	phase- transition backtracking measured (2-SAT $17 \rightarrow 153$, 3-SAT $132 \rightarrow 6646$); $N_c=3$ threshold = Schaefer’s arity-2 bijunctive bound. The EF (Extended Frege) proof- complexity route genuinely EVADES natural proofs and is non-relativizing — a real structural edge over most attempts.	IDENTIFICATION of Schaefer (1978); the curvature framing must produce a prediction Schaefer doesn’t (which new constraint languages are hard) to exceed it

Problem	Status	Barrier (confirmed)	Nearest false neighbor (S2)	What is real	Honest tier
NS	ATTEMPT (pending)	Tao 2016 — an averaged 3D NS that blows up in finite time while obeying the same energy identity & scaling; refutes any energy-monotonicity- <i>only</i> argument	the averaged equation	$\text{Sym}^2(V)/\text{enstrophy}$ structure is a constraint on the <i>algebraic form of the nonlinearity</i> , not a norm — so it <i>may</i> evade Tao. Decisive test: name one step the averaged equation violates.	ATTEMPT — alive IFF the evading step is exhibited
Hodge	~30% PARTIAL	integral Hodge conjecture is FALSE (Atiyah–Hirzebruch 1962) — a proof of the integral version has a bug	the integral conjecture	theta-surjectivity [?] Hodge strategy; $m_s=3=N_c$ kills one non-tempered obstruction; bounded to D_{IV}^5 -type Shimura varieties (BST is one geometry; Hodge is all of them)	PARTIAL — killing one obstruction $\neq$ surjectivity; generalized Kuga–Satake bridge is the open construction. (Don't re-derive divisors/K3s — already done, 0% progress)
4-Color	RE-DERIVATION	K_5 is non-planar — if the argument applies unchanged it proves something false	K_5 (and any non-planar graph)	a genuine re-derivation of a proved theorem — the cleanest, most <i>verifiable</i> win on the list	RE-DERIVATION of Appel–Haken; real mathematics; must survive the K_5 test and not gloss the 4-vs-5 gap (the whole difficulty)
Poincaré	RE-DERIVATION	—	—	re-derivation of Perelman (proved)	RE-DERIVATION; stands
YM	ATTEMPT	the R^4 area-law mass-gap core is a LARGE gap	—	least tractable; honest	ATTEMPT — large open gap

Synthesis (T1276)	META	—	—	common iso-invariant = rank-2 B_2 curvature of D_{IV}^5 ; 15 cross-iso edges	framework recognition, not a proof
----------------------	------	---	---	---	--

Final test verdicts (K1592 — all against pre-registered criteria, none relaxed)

- 4-Color: IDENTIFICATION of the answer, NOT yet a proof. World-confirmed — a conference rejected it with “novel concepts, can’t verify, if correct it’s important, seek critical review,” exactly the K1591 finding (load-bearing steps empirical, not proved). Make-or-break = the interlocking Kempe chains at a degree-5 vertex (Heawood 1890): does T154 (Lemma B) PROVE the interlock resolves, or assert it “2382/2382”? Refine = prove T154 Steps 1/5/6, or carry discharging. Getting the integer $4 \neq$ proving the theorem — must not share a tier line.
- NS: ALIVE, single named gap. The evasion step is REAL — $\int \omega \cdot S \cdot \omega$, the single-field strain–vorticity coupling that Tao’s synthetic $\tilde{B}$ lacks (a property, not provenance; true of B , false of $\tilde{B}$; why 2D can’t blow up and 3D can). But the current draft leans on Tao-shared scaling, not on it; and even re-routed, forcing persistent positive production IS the open NS problem. Better than a dead energy-only argument; below RH.
- Hodge: false-neighbor PASS. Kudla–Millson is automorphic over $\mathbb{Q} \rightarrow$ the RATIONAL conjecture only, torsion-blind, cannot reach the false integral neighbor. Bounded to rational + D_{IV}^5 -Shimura, ~30%.
- $P \neq NP$: IDENTIFICATION of Schaefer (1978) + Karp (1972). Re-deriving a classification geometrically is a consistency check, not progress — a separation needs a lower bound, and classifications are compatible with $P=NP$. Central sentence: “Schaefer classifies; curvature must BOUND — in the unsatisfiable phase.” Wall unmoved.

The three standing rules this ledger is built on

- S1 (family rule): a BST rational form matching one member of a transcendental/analytic family owes a mechanism AND the rest of the family. (Killed 7/8 on Γ -factors and 64/15 on k-SAT thresholds.)
- S2 (nearest-false-neighbor): name the nearest FALSE neighboring statement and run the argument against it; proving the false one is a bug. Five settled false neighbors used (rank-cap/Elkies, K_5 AND K_4 , integral-Hodge, averaged-NS).
- S3 (shared-integer / target-innocence-first): a BST integer reachable by ≥ 2 target-innocent readings owes a UNIQUE forced reading + a forced map before ANY identification — enumerate the readings first. The run’s through-line: color-3 $\neq$ space-3 (reality/parity); the four+ 27s (3^3 , N_c^3 , qqq tensor, 27 lines on a cubic, the real (2,2) irrep); the four 3s in $P \neq NP$ (Schaefer / chromatic / α_c phase-transition / N_c confinement); the historical 6, 3, 2, $SU(2)$. A shared integer is not a shared object.

Honest bottom line

BST has one genuine potential win among the Clay problems if a barrier is cleared (NS, if $\text{Sym}^2(V)$ evades Tao), one clean re-derivation of a proved theorem (4-Color), one real structural edge that doesn’t close the problem ($P \neq NP$ ’s EF route evading natural proofs), and the rest share RH’s single wall. That is a defensible, honest map of the hardest problems in mathematics — an attempt that respects its walls, which is the only kind worth publishing beside real work. No proof is claimed. The Clay problems remain open.

— Keeper, 2026-08-16. Internal. Barriers confirmed (Razborov–Rudich, Aaronson–Wigderson, Tao 2016, Atiyah–Hirzebruch 1962, Elkies 2006, Schaefer 1978, Dokchitser parity). Nothing pushed. CP existence-only.